Energy Decay and Uniqueness for the Heat Equation

An End-to-End Meta-Math Walkthrough

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Abstract

The squared L^2 difference is an energy. Its derivative is the negative of a nonnegative gradient integral, so identical initial states cannot separate. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PDE-03 / Partial Differential Equations / Report v1

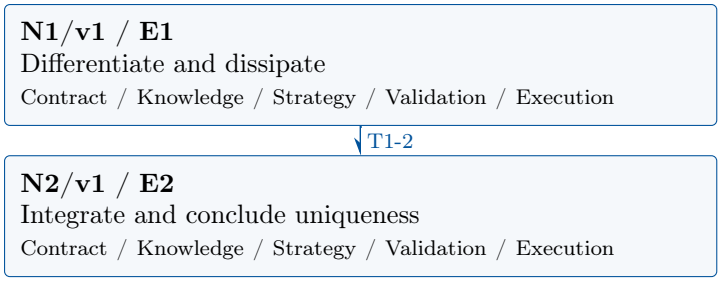
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $\Omega \subset \mathbb{R}^d$ be bounded with C^1 boundary, $T > 0$, and $\kappa > 0$. Suppose real u, v have continuous spatial derivatives through order two and time derivative through order one on $\overline{\Omega} \times [0, T]$, solve $u_t - \kappa \Delta u = v_t - \kappa \Delta v$, and agree on the lateral boundary. Prove $\|u(\cdot, t) - v(\cdot, t)\|_2^2 + 2\kappa \int_0^t \|\nabla(u - v)(\cdot, s)\|_2^2 ds = \|u(\cdot, 0) - v(\cdot, 0)\|_2^2$. In particular equal initial data imply $u = v$.*

A. Foundations and permitted imports

Use differentiation under an integral for continuous derivatives on the compact cylinder, Green's integration-by-parts identity on the C^1 domain for the stated regularity, and the fundamental theorem of calculus. The difference $w = u - v$ solves a homogeneous heat equation with zero boundary values. No existence result or unbounded-domain uniqueness is asserted here.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $E(t) = \int_\Omega w^2$ satisfies $E'(t) = -2\kappa \int_\Omega |\nabla w|^2$.

Receiving activity. N2.1

Compatibility. E1/v1 and nonnegativity of both energy and dissipation. Check: An integral norm equal to zero gives a.e. equality first; continuity supplies equality at every interior point and the boundary data cover the boundary.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Differentiate and dissipate

Object and role. Multiply by the solution difference and transfer one spatial derivative by Green's formula. **Local obligation:** $E(t) = \int_\Omega w^2$ satisfies $E'(t) = -2\kappa \int_\Omega |\nabla w|^2$.

Inputs. $w = u - v$, $w_t = \kappa \Delta w$, and $w|_{\partial\Omega} = 0$.

C — Contract

Publish E1/v1 only after establishing this obligation: $E(t) = \int_\Omega w^2$ satisfies $E'(t) = -2\kappa \int_\Omega |\nabla w|^2$.

K — Knowledge

$w = u - v$, $w_t = \kappa \Delta w$, and $w|_{\partial\Omega} = 0$. Retain the stated assumptions and import versions.

S — Strategy

Multiply by the solution difference and transfer one spatial derivative by Green's formula.

V — Validation

The boundary term vanishes because $w = 0$, not because its normal derivative is assumed zero.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 The regularity and compactness allow $E'(t) = 2\int_\Omega w w_t = 2\kappa \int_\Omega w \Delta w$.

N1.2 Green's identity gives $\int_\Omega w \Delta w = \int_{\partial\Omega} w \partial_\nu w - \int_\Omega |\nabla w|^2$. The boundary value of w is zero, so the boundary integral vanishes. This proves the exact energy derivative.

Export E1/v1. $E(t) = \int_\Omega w^2$ satisfies $E'(t) = -2\kappa \int_\Omega |\nabla w|^2$.

Acceptance gate. The boundary term vanishes because $w = 0$, not because its normal derivative is assumed zero.

Targeted repair. If the boundary term is simply omitted, write it explicitly and use the equality of boundary data to make the difference vanish there. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Integrate and conclude uniqueness

Object and role. Integrate the identity from zero to t, then use continuity to upgrade an L^2 zero to pointwise equality. **Local obligation:** The stated energy identity holds, and equal initial data give equality of the two classical solutions.

Inputs. E1/v1 and nonnegativity of both energy and dissipation. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: The stated energy identity holds, and equal initial data give equality of the two classical solutions.

K — Knowledge

E1/v1 and nonnegativity of both energy and dissipation. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Integrate the identity from zero to t, then use continuity to upgrade an L^2 zero to pointwise equality.

V — Validation

An integral norm equal to zero gives a.e. equality first; continuity supplies equality at every interior point and the boundary data cover the boundary.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Integrating E1 yields $E(t) + 2\kappa \int_0^t \int_\Omega |\nabla w(x,s)|^2 \, dx \, ds = E(0)$. This proves the claimed stability identity and $E(t) \le E(0)$.

N2.2 If the initial data agree, $E(0) = 0$. Nonnegativity forces $E(t) = 0$ for each t. Were continuous w nonzero at an interior point, it would be bounded away from zero on a small ball, making the integral positive. Thus $w = 0$ inside, and it is zero on the boundary by assumption.

Export E2/v1. The stated energy identity holds, and equal initial data give equality of the two classical solutions.

Acceptance gate. An integral norm equal to zero gives a.e. equality first; continuity supplies equality at every interior point and the boundary data cover the boundary.

Targeted repair. If equality almost everywhere is treated as pointwise without regularity, use continuity explicitly or state uniqueness only in the appropriate equivalence class. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Differentiate and dissipate

1. **N1.1.** The regularity and compactness allow $E'(t) = 2\int_\Omega w w_t = 2\kappa \int_\Omega w\Delta w$.
2. **N1.2.** Green's identity gives $\int_\Omega w\Delta w = \int_{\partial\Omega} w\partial_\nu w - \int_\Omega |\nabla w|^2$. The boundary value of w is zero, so the boundary integral vanishes. This proves the exact energy derivative.

B. N2 — Integrate and conclude uniqueness

1. **N2.1.** Integrating E1 yields $E(t) + 2\kappa \int_0^t \int_\Omega |\nabla w(x,s)|^2\, dx\, ds = E(0)$. This proves the claimed stability identity and $E(t) \le E(0)$.
2. **N2.2.** If the initial data agree, $E(0) = 0$. Nonnegativity forces $E(t) = 0$ for each t. Were continuous w nonzero at an interior point, it would be bounded away from zero on a small ball, making the integral positive. Thus $w = 0$ inside, and it is zero on the boundary by assumption.

□

V. WORKED ORCHESTRATION

On $(0,\pi)$ with $\kappa = 1$, take $u(x,t) = e^{-t}\sin x$ and $v = 0$. Then $E(t) = (\pi/2)e^{-2t}$, and $\|u_x(\cdot,t)\|_2^2 = (\pi/2)e^{-2t}$. Hence $E(t) + 2\int_0^t \|u_x\|_2^2 ds = \pi/2 = E(0)$, checking the constants and sign.

A. Boundary and hypothesis audit

If boundary equality is removed, $u(x,t) = x$ and $v = 0$ on $(0,1)$ are both time-independent heat solutions. Their difference has constant energy $1/3$, while its gradient integral is one. The missing boundary flux equals one and exactly cancels the dissipation term; it cannot be discarded.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 identity: $E' = -2\kappa\|\nabla w\|_2^2$ for arbitrary boundary data. **Validation witness:** $w = x$ on $(0,1)$. **Repair:** retain the flux term in Green's identity and then use the

actual homogeneous difference boundary condition from §1. Recheck the integrated identity and uniqueness. This repairs the proof of the stated class, not a broader uniqueness claim.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Extend the energy method to weak parabolic solutions by justifying time testing through approximation.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Leon Simon, Lectures on PDE, for elliptic, weak, and parabolic methods. Classical calculus and the explicitly stated Hilbert-space imports are used below; the displayed PDE arguments are developed in full.